

141. Let $\alpha = 8 - 14i$, $A = \left\{ z \in \mathbb{C} : \frac{\alpha z - \bar{\alpha} \bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1 \right\}$ and $B = \{ z \in \mathbb{C} : |z + 3i| = 4 \}$. Then $\sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z)$ is equal to _____.

माना $\alpha = 8 - 14i$, $A = \left\{ z \in \mathbb{C} : \frac{\alpha z - \bar{\alpha} \bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1 \right\}$ तथा $B = \{ z \in \mathbb{C} : |z + 3i| = 4 \}$ है। तो $\sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z)$ बराबर है।

Ans. 14

Sol. $\alpha = 8 - 14i$

$$z = x + iy$$

$$\alpha z = (8x + 14y) + i(-14x + 8y)$$

$$z + \bar{z} = 2x \quad z - \bar{z} = 2iy$$

$$\text{Set A : } \frac{2i(-14x + 8y)}{i(4xy - 112)} = 1$$

$$(x - 4)(y + 7) = 0$$

$$x = 4 \quad \text{or} \quad y = -7$$

$$\text{Set B : } x^2 + (y + 3)^2 = 16$$

$$\text{when } x = 4 \quad y = -3$$

$$\text{when } y = -7 \quad x = 0$$

$$\therefore A \cap B = \{4 - 3i, 0 - 7i\}$$

$$\text{So, } \sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z) = 4 - (-3) + (0 - (-7)) = 14$$

142. A boy needs to select five courses from 12 available course, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is _____.

12 उपलब्ध पाठ्यक्रमों, जिनके 5 भाषा के पाठ्यक्रम हैं, में से एक लड़के को पाँच पाठ्यक्रम लेने हैं। यदि वह अधिकतम दो भाषा के पाठ्यक्रम ले सकता है, तो उसके द्वारा पाँच पाठ्यक्रम लेने के तरीकों की संख्या है।

Ans. 546

Sol. For at most two language courses

$$= {}^5C_2 \times {}^7C_3 + {}^5C_1 \times {}^7C_4 + {}^7C_5 = 546$$

143. If the coefficient of x^{30} in the expansion $\left(1 + \frac{1}{x}\right)^6 (1 + x^2)^7 (1 - x^3)^8$; $x \neq 0$ is α , then $|\alpha|$ equals _____.

यदि $\left(1 + \frac{1}{x}\right)^6 (1+x^2)^7 (1-x^3)^8$; $x \neq 0$, के प्रसार में x^{30} का गुणांक α है, तो $|\alpha|$ बराबर है _____

Ans. 678

Sol. coefficient of x^{30} in $\frac{(x+1)^6 (1+x^2)^7 (1-x^3)^8}{x^6}$

coefficient of x^{36} in $(1+x)^6 (1+x^2)^7 (1-x^3)^8$

General term

$${}^6C_{r_1} {}^7C_{r_2} {}^8C_{r_3} (-1)^{r_3} x^{r_1+2r_2+3r_3}$$

$$r_1 + 2r_2 + 3r_3 = 36$$

Case – I :

r_1	r_2	r_3
0	6	8
2	5	8
4	4	8
4	3	8

$$r_1 + 2r_2 = 12 \text{ (taking } r_3 = 8)$$

Case – II :

r_1	r_2	r_3
1	7	7
3	6	7
5	5	7

$$r_1 + 2r_2 = 15 \text{ (taking } r_3 = 7)$$

Case – III :

r_1	r_2	r_3
4	7	6
6	6	6

$$r_1 + 2r_2 = 18 \text{ (taking } r_3 = 6)$$

$$\text{Coefficient} = 7 + (15 \times 21) + (15 \times 35) + (35)$$

$$-(6 \times 8) - (20 \times 7 \times 8) - (6 \times 21 \times 8) + (15 \times 28)$$

$$+ (7 \times 28) = -678 = \alpha$$

$$|\alpha| = 678$$